

Ex 1. Tonks' Gas in 1 dimension.

Suppose we have N slender rods of length R in a "volume" L ($L \gg NR$) so that we are in the diluted limit (ideal gas limit). Find the equation of state of the system using the microcanonical example. The rods do not interact — only excluded volume.

General approach:

1. compute $\mathcal{S} = k_B \ln \Omega$

$$\Omega = \int dx \int dp \delta(E - H(x, p))$$

2. compute T and P according to the relationships

$$\frac{1}{T} = \left(\frac{\partial \mathcal{S}}{\partial E} \right)_{V, N} \quad \frac{P}{T} = \left(\frac{\partial \mathcal{S}}{\partial V} \right)_{E, N}$$

1. We have separate integrals for positions and momenta.

For momenta, we get the same as for the ideal gas, except for the fact that we have one moment coordinate p_x therefore $\Omega(p) \equiv$ volume of $N-1$ hypersphere of radius $\sqrt{2mE}$

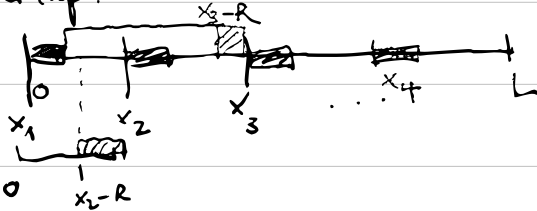
This means that since

$$V^{l/2}(R) = \frac{\pi^{l/2}}{\Gamma(\frac{l}{2}+1)} R^l \quad V^N(R) = \frac{\pi^{N/2}}{\Gamma(\frac{N}{2}+1)} (2mE)^{N/2}$$

The moments in a slice $E, E+\delta E$

$$\frac{d\Omega(E)}{dE} = \frac{\pi^{N/2}}{\Gamma(\frac{N}{2}+1)} (2m)^{N/2} \frac{N}{2} E^{\frac{N}{2}-1}$$

Then for the partitioning we have excluded volume effects since the slender rods cannot overlap:



$$\int_0^{x_2-R} dx_1 = x_2 - R$$

$$\int_R^{x_3-R} dx_2 (x_2 - R) = \left. \frac{(x_2 - R)^2}{2} \right|_R^{x_3-R} = \frac{(x_3 - 2R)^2}{2}$$

$$\int_{2R}^{x_4-R} dx_3 \frac{(x_3 - 2R)^2}{2} = \left. \frac{(x_3 - 2R)^3}{6} \right|_{2R}^{x_4-R} = \frac{(x_4 - 3R)^3}{6}$$

$$\dots$$

$$\int_{(N-1)R}^{L-R} dx_N \frac{(x_N - (N-1)R)^{N-1}}{(N-1)!} = \left. \frac{(x_N - (N-1)R)^N}{N!} \right|_{(N-1)R}^{L-R} = \frac{(L - NR)^N}{N!}$$

touching rod 1

Putting everything together we have that:

$$\Omega(E) = \frac{(L-NR)^N}{N!} \frac{(2mE\pi)^{N/2}}{\Gamma(N/2+1)} \frac{N}{2E}$$

$$S = k_B \ln \Omega(E) \underset{\ln x \approx N}{\approx} N k_B \ln \left[\frac{(L-NR)(2mE\pi)^{1/2}}{N^{3/2}} \right] + k_B \frac{3N}{2}$$

from here we have that

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E} \right)_{N,V} = \left(\frac{N}{2} - 1 \right) \cdot \frac{1}{E} k_B$$

$$\frac{P}{T} = \left(\frac{\partial S}{\partial L} \right)_{N,E} = N \frac{k_B}{L-NR} \rightarrow$$

$$\boxed{P(L-NR) = N \cdot T k_B}$$

↓
accessible volume

ideal gas law
as expected

(In class) - Microcanonical energy function

from the analysis of 2 subsystems with E_1, E_2 $E_1 + E_2$ we know that

$$\Omega(E_1) = \Omega(E_1^*) \Omega(E_2^*) e^{-\frac{1}{2} \left(\frac{\partial S}{\partial E_1} + \frac{\partial S}{\partial E_2} \right) (E_1 - E_1^*)} k_B$$

(a) Show that $\left(\frac{1}{k_B} \frac{d^2 S}{dE^2} \right)_{N,V} = - \frac{1}{k_B T} \frac{1}{N C_V}$

where $\left(\frac{\partial E}{\partial T} \right)_{N,V} = N C_V$

Notice that $\frac{\partial^2 S}{\partial E^2} = \frac{\partial}{\partial E} \left(\frac{\partial S}{\partial E} \right) = \frac{\partial}{\partial T} \left(\frac{1}{T} \right) \frac{\partial T}{\partial E} \Big|_{N,V}$

$$= - \frac{1}{T^2} \frac{1}{\left(\frac{\partial T}{\partial E} \right)_{N,V}} = - \frac{1}{T^2} \frac{1}{N C_V}$$

So that $\left(\frac{1}{k_B} \frac{d^2 S}{dE^2} \right)_{N,V} = - \frac{1}{k_B T} \frac{1}{N C_V}$

and $\sigma_E^2 = - \frac{1}{k_B \left(\frac{\partial^2 S}{\partial E^2} \right)_{N,V}} = T^2 N C_V$

$\sigma_E \sim \sqrt{N}$ if we look at E/N
Energy per particle then

$$\sigma_{E/N} \sim \frac{1}{\sqrt{N}}$$

→ Note that this shows that we can learn the specific heat by observing energy fluctuations - Fluctuation-dissipation relationship

b) If $\omega^{(1)}$ and $\omega^{(2)}$ are the specific heats of two subsystems show that

$$\frac{1}{\omega^1} + \frac{1}{\omega^2} = \frac{N k_B T^2}{\sigma_E^2}$$

from before, we have that

$$\sigma_E^2 = - \frac{K_B}{\left(\frac{\partial S}{\partial E} \right)_{N,V} + \frac{\partial S_2}{\partial E_2}} =$$

$$\left(\frac{\partial S_1}{\partial E_1} \right)_{N,V} = - \frac{1}{T^2 N \omega^{(1)}} \quad \left(\frac{\partial S_2}{\partial E_2} \right)_{N,V} = - \frac{1}{T^2 N \omega^{(2)}}$$

so that

$$\sigma_E^2 = \frac{1}{T^2 N} \cdot \frac{K_B}{\left(\frac{1}{\omega^1} + \frac{1}{\omega^2} \right)}$$

$$\boxed{\frac{1}{\omega^{(1)}} + \frac{1}{\omega^{(2)}} = \frac{T^2 N K_B}{\sigma_E^2}} \quad 0$$

and $\sigma_E \equiv 0$

c) assuming $K = \langle E_1 \rangle \rightarrow$ mean energy.

The equipartition theorem says that

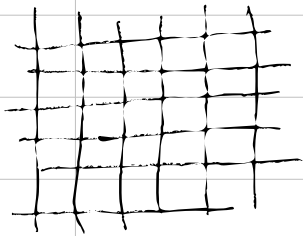
$$K = \frac{1}{2} k_B T \times \text{degree of freedom} : 3N$$

$$= \frac{3}{2} k_B T N = T \frac{\frac{3}{2} k_B}{N}$$

The "fluctuation-dissipation" says:

$\downarrow$ sum of terms. (only k contribution) $\rightarrow E = \frac{3}{2} k_B T N$
 $\alpha = \left(\frac{1}{N} \frac{\partial E}{\partial T} \right)_{N,N} = \frac{3}{2} k_B N \frac{1}{N} = \frac{3}{2} k_B \quad \checkmark$

Exercise 5 in collection: Schottky vacancies



lattice N sites

n sites are empty

$$E = \epsilon n$$

$$\Omega(E) = \binom{N}{N-n} \quad \text{combinations}$$

$$S = k_B \ln \Omega(E) = k_B \ln \frac{N!}{(N-n)! n!}$$

$$= k_B [N(\ln N - 1) - (N-n)(\ln(N-n) - 1) - n(\ln n - 1)]$$

N large

$$= k_B [N \ln N - (N-n) \ln(N-n) - n \ln n - \cancel{N} + \cancel{N-n} + \cancel{n}]$$

$$= N k_B \left[\ln N - \left(1 - \frac{n}{N}\right) \left[\ln N + \ln \left(1 - \frac{n}{N}\right) \right] - \frac{n}{N} \ln n \right]$$

$$= N k_B \left[\ln \cancel{N} - \cancel{N} + \frac{n}{N} \ln N - \left(1 - \frac{n}{N}\right) \ln \left(1 - \frac{n}{N}\right) - \frac{n}{N} \ln n \right]$$

$$\tilde{N} = -N k_B [(1-x) \ln(1-x) + x \ln x] \quad (\geq 0)$$

$x = \frac{n}{N} = \frac{E}{2N}$
 $\rightarrow \frac{1}{T} = \left(\frac{\partial S}{\partial E} \right)_{N,N} = \frac{\partial S}{\partial x} \cdot \frac{\partial x}{\partial E} \stackrel{\approx \frac{1}{2}N}{=} -N k_B \left[\ln(1-x) + \ln x \right] \frac{1}{2N}$
 $\quad \quad \quad = -\frac{N k_B}{2N} \ln \frac{x}{1-x}$

$$\frac{1}{T} = - \frac{k_B}{r} \ln \frac{x}{1-x}$$

$$\rightarrow \frac{x}{1-x} = e^{-r/k_B T} \quad \frac{n}{N} = x = \frac{e^{-r/k_B T}}{1 + e^{-r/k_B T}}$$

$$\text{so } E = n_f = rN \frac{e^{-r/k_B T}}{1 + e^{-r/k_B T}}$$

$$\text{for } T \rightarrow 0 \quad x \rightarrow 0 \quad ; \quad E \rightarrow 0$$

$$S = \lim_{x \rightarrow 0} -k_B N [x \ln x + (1-x) \ln(1-x)]$$

$$b) \text{ For } C_V = \left(\frac{1}{N} \frac{\partial E}{\partial T} \right)_{N,P}$$

we get that

$$\begin{aligned} \frac{\partial E}{\partial T} &= rN \left[\frac{e^{-r/k_B T} \cdot \frac{r}{k_B T^2}}{1 + e^{-r/k_B T}} - \frac{(e^{-r/k_B T})^2 \frac{r}{k_B T^2}}{(1 + e^{-r/k_B T})^2} \right] \\ &= \frac{r^2 N}{k_B T^2} [x - x^2] \end{aligned}$$

$$C_V = \left(\frac{r}{T} \right)^2 \frac{1}{k_B} [x - x^2] \geq 0 \leftarrow \max x = \frac{1}{2}$$



Ex. 3.

N non-interacting harmonic oscillators in 3D

$$H_i = \frac{p_i^2}{2m} + \frac{1}{2} m \omega^2 q_i^2 \quad H = \sum H_i$$

$$\Omega(E) = \int \frac{dQ dP}{h^{3N}} \delta \left(E - \sum_i \left(\frac{p_i^2}{2m} + \frac{1}{2} m \omega^2 q_i^2 \right) \right)$$

$$\Rightarrow \text{change of variable} \quad x_i = \frac{p_i}{\sqrt{2m}} \quad y_i = \frac{m\omega}{\sqrt{2}} q_i$$

$$\left\{ \begin{aligned} dp_i &= \sqrt{2m} dx_i \\ dq_i &= \sqrt{\frac{2}{m\omega}} dy_i \end{aligned} \right\} \quad \delta \left(E - \sum_i x_i^2 + y_i^2 \right)$$

↓
GN sphere of radius $\sqrt{E}$

$$\rightarrow dp_i dq_i = \sqrt{\frac{4m}{\mu}} \frac{1}{\omega} dx_i dy_i = \frac{2}{\omega} dx_i dy_i$$

Volume of this sphere will be

$$V_n(R) = \frac{\pi^{n/2} R^n}{\Gamma(\frac{n}{2} + 1)} \rightarrow \text{Surface take the dens}$$

with: $n = 6N$ $R = \sqrt{E}$

$$\Omega(E) = \frac{1}{h^{3N}} \left(\frac{2}{\omega} \right)^{3N} \int d\mathbf{x} d\mathbf{y} \delta(E - \sum_i x_i^2 + y_i^2)$$

↓ surface of the energy shell

$$\Omega(E) = \left(\frac{2}{\omega} \right)^{3N} \frac{\partial}{\partial E} \left(\frac{(\pi E)^{6N/2}}{(\frac{6N}{2} + 1)!} \right) \frac{1}{h^{3N}} \leftarrow \text{Question where is the volume in here?}$$

$$= \left(\frac{2}{\omega} \right)^{3N} \frac{(\pi E)^{3N}}{(3N+1)!} \cdot \frac{3N}{E} \frac{1}{h^{3N}}$$

$$S = k_B \ln \Omega \approx 3N k_B \left[\log \left[\frac{2\pi E}{h\omega^{3N}} \right] + 1 \right]$$

for larger

$$\left(\frac{\partial S}{\partial E} \right)_{N,V} = 3N \frac{k_B}{E} = \frac{3N k_B}{E} = \frac{1}{T} \quad \boxed{E = 3N k_B T}$$

→ Note what happens if they are indistinguishable?

S is no longer extensive! $\Omega' = \Omega/N!$

$$S \approx 3N k_B \left[\log \left[\frac{2\pi E}{h\omega^{3N}} \right] + 1 - \log N^{1/3} + \frac{1}{3} \right]$$

$$3N k_B \left[\log \left(\frac{2\pi E}{h\omega^{3N^{4/3}}} \right) + \frac{4}{3} \right]$$

Since $E \propto N$ $S \propto 3N \log N^{-1/3}$

(Ex4.) Quantum oscillators.

$$E_n = \hbar \omega \left(n + \frac{1}{2} \right)$$

$$E = \sum_{n=1}^{3N} E_n = \sum_{n=1}^{3N} \left(n + \frac{1}{2} \right) \hbar \omega$$

$$\sum n_i = \frac{E}{\hbar \omega} - \frac{3N}{2} \equiv M$$

when the condition $\sum_{i=1}^{3N} m_i = M$ holds
 the # of possible configurations for the
 m_i 's are: $\binom{M+3N-1}{M} = \frac{(M+3N-1)!}{M! (3N-1)!}$

$$= \frac{(E/k_B T + 3N/2 - 1)!}{(E/k_B T - \frac{3N}{2})! (3N-1)!}$$

$$S = k_B \ln \Omega(E, N)$$

for large $E/k_B T, N$

$$\ln(E/k_B T + \frac{3N}{2} - 1)! \sim (E/k_B T + \frac{3N}{2} - 1) \left[\ln(E/k_B T + \frac{3N}{2} - 1) \right]$$

$$\ln(E/k_B T - \frac{3N}{2})! \sim (E/k_B T - \frac{3N}{2}) \left[\ln(E/k_B T - \frac{3N}{2}) - 1 \right]$$

$$\ln(3N-1)! \sim (3N-1) \left[\ln(3N-1) - 1 \right]$$

Indep Term $- E/k_B T - \frac{3N}{2} + 1 + E/k_B T - \frac{3N}{2} + \frac{3N}{2} - 1$

Other terms $\sim (E/k_B T + \frac{3N}{2}) \left[\ln(E/k_B T + \frac{3N}{2}) \right]$

$$- (E/k_B T - \frac{3N}{2}) \left[\ln(E/k_B T - \frac{3N}{2}) \right]$$

$$- 3N \ln 3N = N \left[(E/k_B T + \frac{3}{2}) \ln(E/k_B T + \frac{3}{2}) - (E/k_B T - \frac{3}{2}) \ln(E/k_B T - \frac{3}{2}) \right. \\ \left. + (E/k_B T + \frac{3}{2} - E/k_B T + \frac{3}{2}) \ln N - 3 \ln N \right]$$

$$S \approx k_B N \left[\left(\frac{E}{k_B T N} + \frac{3}{2} \right) \ln \left(\frac{E}{k_B T N} + \frac{3}{2} \right) \right. \\ \left. - \left(\frac{E}{k_B T N} - \frac{3}{2} \right) \ln \left(\frac{E}{k_B T N} - \frac{3}{2} \right) \right]$$

$$x = \frac{E}{k_B T N} \quad S = k_B N \left[(x + 3/2) \ln(x + 3/2) \right. \\ \left. - (x - 3/2) \ln(x - 3/2) \right]$$

$$\frac{\partial S}{\partial E} = \frac{\partial S}{\partial x} \frac{\partial x}{\partial E} = \frac{N k_B}{k_B T N} \left[\ln \frac{x+3/2}{x-3/2} + 1 - 1 \right]$$

$$\frac{1}{T} = \frac{N k_B}{k_B T N} \ln \frac{x+3/2}{x-3/2} \quad \frac{x+3/2}{x-3/2} = e^{\frac{k_B T}{k_B T}}$$

$$x + \frac{3}{2} = e^{\frac{k_B T}{k_B T}} (x - \frac{3}{2}) \rightarrow x = \frac{\frac{3}{2} (1 + e^{\frac{k_B T}{k_B T}})}{1 - e^{\frac{k_B T}{k_B T}}}$$

$$E = \frac{3}{2} k_B T N \ln \left(\frac{k_B T}{2 k_B T} \right)$$

$$E = \frac{3}{2} N \hbar \omega \coth \left(\frac{\hbar \omega}{2 k_B T} \right)$$

Note that for $k_B T \gg \hbar \omega$

$$\frac{e^x + e^{-x}}{e^x - e^{-x}} \xrightarrow{x \rightarrow 0} \frac{2}{2x} \sim \frac{1}{x}$$

$$E \rightarrow \frac{3}{2} N \hbar \omega \cdot \frac{1}{\frac{\hbar \omega}{2 k_B T}} = 3 N k_B T$$

↑
answer for
classical
oscillators
in Ex #3.

Ex. 6

Identical particles — n_+ = # part with energy ϵ
 n_- = # part with energy $-\epsilon$

$$\rightarrow E = n_+ \epsilon - n_- \epsilon = (2n_+ - N) \epsilon$$

$$n_+ + n_- = N \quad \rightarrow \quad n_+, n_- \text{ occupation numbers.}$$

$$\rightarrow n_+ = \left(\frac{E}{\epsilon} + N \right) \frac{1}{2}$$

$$\Omega(E) = \binom{N}{n_+} = \frac{N!}{n_+! (N-n_+)!}$$

$$S = k_B \ln \Omega(E) \approx k_B (N \ln N - n_+ \ln n_+ - (N-n_+) \ln (N-n_+))$$

$N, n_+, N-n_+ \text{ large}$

$$S \approx k_B (N \ln N - n_+ \ln n_+ - (N-n_+) \ln (N-n_+))$$

if we define $x \equiv \frac{n_+}{N}$ we have that

$$\begin{aligned} S &\approx k_B N (\ln N - x \ln x N - (1-x) \ln (1-x) \cdot N) \\ &= k_B N (\ln N - x \ln x - x \ln N - (1-x) \ln (1-x) - (1-x) \ln N) \\ &= -k_B N (x \ln x + (1-x) \ln (1-x)) \end{aligned}$$

Using that $\left. \frac{\partial S}{\partial E} \right|_N = \frac{1}{T} = \frac{\partial S}{\partial x} \frac{\partial x}{\partial E}$

$$\frac{1}{T} = \frac{1}{2\epsilon N} (-k_B N) \left[x \frac{1}{x} + \ln x + (1-x) \frac{1}{1-x} (-1) - \ln(1-x) \right]$$

$$\frac{1}{T} = \frac{k_B}{2\epsilon} \ln \frac{1-x}{x} \rightarrow \frac{1-x}{x} = e^{2\epsilon/k_B T}$$

$$\rightarrow x = \frac{1}{1 + e^{2\epsilon/k_B T}} \rightarrow n_+ = N / (1 + e^{2\epsilon/k_B T})$$

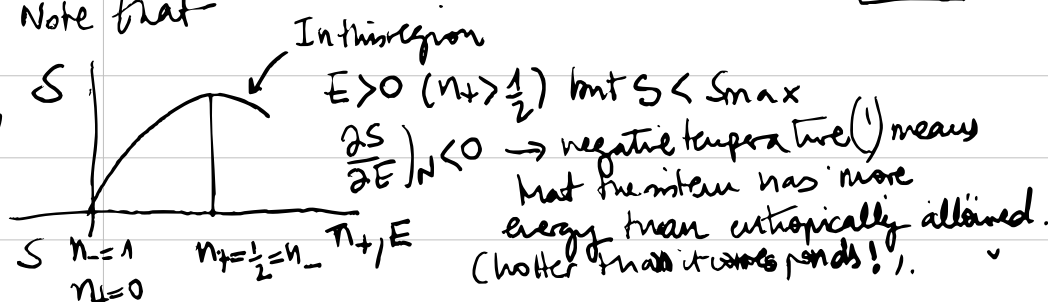
Note that $n_+ = \frac{N}{1+e^{2\epsilon/k_B T}}$ $n_- = \frac{N e^{2\epsilon/k_B T}}{1+e^{2\epsilon/k_B T}}$

$$\begin{aligned} E &= \epsilon (2n_+ - N) = \epsilon N \left[\frac{2 - 1 - e^{2\epsilon/k_B T}}{e^{2\epsilon/k_B T} + 1} \right] \\ &= \epsilon N \left[\frac{1 - e^{2\epsilon/k_B T}}{e^{2\epsilon/k_B T} + 1} \right] = \frac{\epsilon N e^{-\epsilon/k_B T}}{e^{\epsilon/k_B T} + e^{-\epsilon/k_B T}} \\ &= \epsilon N \frac{e^{-\epsilon/k_B T} - e^{+\epsilon/k_B T}}{e^{\epsilon/k_B T} + e^{-\epsilon/k_B T}} = -\epsilon N \tanh\left(\frac{\epsilon}{k_B T}\right) \end{aligned}$$

For $T \rightarrow 0$ $n_+ \rightarrow 0$ $n_- \rightarrow 1$ $E \rightarrow -\epsilon N$ $S \rightarrow 0$

For $T \rightarrow \infty$ $n_+ \rightarrow 1/2$ $n_- \rightarrow 1/2$ $E \rightarrow 0$ $S \rightarrow k_B N \ln 2$

Note that



Free energy: $F = E - TS = -\epsilon N \tanh(\epsilon/k_B T) + T k_B N [x \ln x + (1-x) \ln(1-x)]$

$$x \ln x = \frac{1}{1+e^{2\epsilon/k_B T}} (\ln 1 + e^{2\epsilon/k_B T})$$

$$(1-x) \ln(1-x) = \frac{e^{2\epsilon/k_B T}}{1+e^{2\epsilon/k_B T}} \left[\left[\frac{2\epsilon}{k_B T} \right] - \ln[1+e^{2\epsilon/k_B T}] \right]$$

$$F = -\epsilon N \tanh(\epsilon/k_B T) + T k_B N \frac{2\epsilon}{k_B T} \frac{e^{2\epsilon/k_B T}}{1+e^{2\epsilon/k_B T}} - \square$$

$$\Delta = \frac{e^{2\epsilon/k_B T} - 1}{1+e^{2\epsilon/k_B T}} (\epsilon N) + 2\epsilon N \frac{e^{2\epsilon/k_B T}}{1+e^{2\epsilon/k_B T}}$$

$$\Delta = \epsilon N \frac{1+e^{2\epsilon/k_B T}}{1+e^{2\epsilon/k_B T}}$$

$$\square = -T k_B N \ln(1+e^{2\epsilon/k_B T})$$

$$F = \epsilon N - T k_B N \ln(1+e^{2\epsilon/k_B T})$$

which for $T \rightarrow 0$ $F \approx \epsilon N - T k_B N \frac{2\epsilon}{k_B T} \rightarrow -\epsilon N$ as expected.

7. N particles $\rightarrow$ 2 energy levels
 $+E, 0$
 n_E, n_0 $E = n_E \cdot E$
 $\rightarrow g$ fold degeneracy $n_0 = N - n_E$

But there are g states with energy E
 or that

$$\Omega(E) = \binom{N}{n_0} g^{n_E} \rightarrow \text{each particle } n_E \text{ has probability } \frac{1}{g} \text{ of being in each level.}$$

As in exercise 6 we will get

$$S = -k_B N (x \ln x + (1-x) \ln (1-x)) + \underbrace{k_B N x \ln g}_{\text{ext. term for degeneracy}}$$

$x = \frac{n_E}{N}$

$$\left(\frac{\partial S}{\partial E} \right)_N = \frac{1}{T} = \frac{\partial S}{\partial x} \frac{\partial x}{\partial E} = \frac{1}{EN} (-k_B N) \frac{\ln x}{1-x} + \frac{k_B N \ln g}{EN}$$

$\rightarrow \frac{n_E}{N} = E/EN$

$$\frac{1}{T} = + \frac{k_B}{E} \ln \frac{1-x}{x} + \frac{k_B}{E} \ln g$$

$$\ln \frac{1-x}{x} = \frac{E}{k_B T} - \ln g; \quad \frac{1-x}{x} = e^{E/k_B T} \cdot \frac{1}{g}$$

$$1-x = \frac{x}{g} e^{E/k_B T} \quad x = \frac{1}{1 + e^{E/k_B T} \cdot \frac{1}{g}} = \frac{e^{-E/k_B T} g}{1 + e^{-E/k_B T} g}$$

$$n_E = \frac{e^{-E/k_B T} g}{1 + e^{-E/k_B T} g} \cdot N \rightarrow \text{Boltzmann weight.}$$

$$n_0 = N - n_E = N \frac{1}{1 + e^{-E/k_B T} g}$$

$$E = n_E \cdot E = N E \cdot \frac{e^{-E/k_B T} g}{1 + e^{-E/k_B T} g}$$

for $T \rightarrow 0$ $n_E \rightarrow 0$ $E \rightarrow 0$ $S \rightarrow 0$
 $n_0 \rightarrow 1$

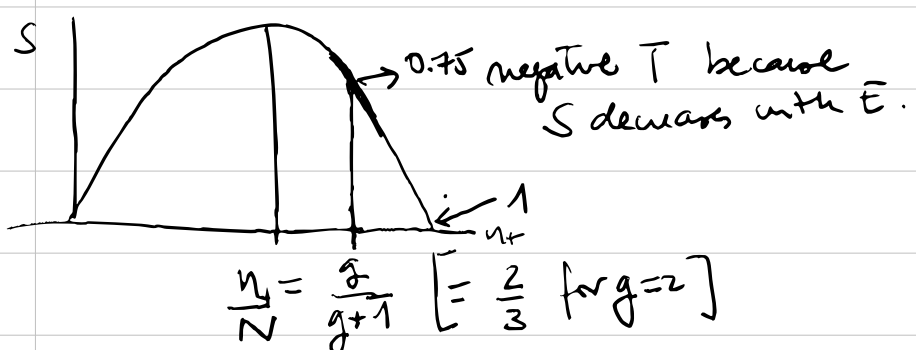
for $T \rightarrow \infty$ $n_E \rightarrow \frac{g}{1+g}$ $E = N E \frac{g}{1+g}$
 $n_0 \rightarrow \frac{1}{1+g}$

b) $E = 0.75 N E$ $x = 0.75$

If $g=2$ $\frac{1}{T} = \frac{k_B}{E} \ln \frac{1/4}{3/4} + \frac{k_B}{E} \ln 2$

$$\frac{1}{T} = \frac{k_B}{E} [-\ln 3 + \ln 2] < 0!$$

In fact, from the result in a).



2 we put the system in contact with a heat bath

$$\begin{aligned} \Delta S &= \Delta S_{\text{bath}} + \Delta S_{\text{system}} \\ &= \left(\left(\frac{\partial S}{\partial E} \right)_b - \left(\frac{\partial S}{\partial E} \right)_s \right) \cdot \Delta E \quad \leftarrow \text{total energy that flows from system to bath.} \\ &= \left(\frac{1}{T_{\text{bath}}} - \frac{1}{T_{\text{system}}} \right) \Delta E \geq 0 \end{aligned}$$

$\underbrace{\hspace{10em}}_{>0}$ $\downarrow$
has to be >0 for $\Delta S_{\text{system}} > 0$

so that ΔE flows from system to heat bath or "negative temp" means the system is "too hot" and needs to lose energy to be at equilibrium.

